Question 1: How do you decide that an Ethernet frame is targeting your machine?
I decide if a frame is for my machine by checking its destination MAC address. If the MAC matches one of the physical addresses I loaded from my ARP mapping file, then it belongs to one of my virtual interfaces. I also accept any frames with the broadcast address ff:ff:ff:ff:ff:ff since those are intended for everyone on the local network.

Question 2: Next, how do you determine that it's an ARP request asking about your machine's MAC address?
Once I know the frame is for me (or is a broadcast), I check the Ethernet Type field to see if it's 0x0806, which identifies it as an ARP packet. If it's ARP, I look at the operation field to make sure it's an ARP Request (value 1). Then, I compare the "Target Protocol Address" in the ARP message with my own set of assigned IP addresses to see if anyone is asking specifically about me.

Question 3: How do you determine it's a proper ICMP echo request targeting your machine?
For ICMP, I first check that the Ethernet Type is 0x0800 (IPv4) and then verify that the IP Protocol byte is 1 (ICMP). I make sure the destination IP address in the IP header is one of mine. If these both of these are true, then I look at the first byte of the ICMP header; if it's type 8, I know it's an Echo Request.

Question 4: Which fields of the outgoing packet's IP header need to be modified?
To generate the reply, I have to swap the source and destination IP addresses so the packet goes back to the requester. I also update the Identification field using a sequential counter that starts at 1000 and increments for every reply. For the Header Checksum, I reset it to zero and then recalculate it because the IPs and ID have changed. In mine, I decided to preserve the Flags and fragmentation offset from the original request to maintain consistency with the incoming traffic, which helped in to pass the tests.

Question 5: Do you need to recalculate the checksum of the IP header? of the ICMP message? or both?
Both checksums need to be recalculated. The IP header checksum changes because I've modified the source/destination IPs and the Identification field. The ICMP checksum also changes because I change the Type field from 8 (Request) to 0 (Reply). I used a helper function to do both of these calculations by adding the 16-bit words and doing the one's complement.